

Agenda

- Create a table in MySQL
- Practical Example of Foreign Keys

→ CRUD

→ Create ✓

→ Read

→ DISTINCT

→ PRINT constant value

→ Operations on Columns

→ Inserting data from Another Table

→ WHERE clause

→ AND, OR, NOT

→ IN operator

→ BETWEEN operator

→ LIKE

→ IS NULL

→ ORDER BY

→ LIMIT

→ UPDATE

→ DELETE

→ Delete vs Truncate vs Drop

→ Truncate

→ Drop

* You can attempt CRUD
Assignments after today's
class.

Create table in MySQL

SQL
→ Structured Query Language

```
CREATE TABLE students (  
  id INT AUTO_INCREMENT,   
  firstName VARCHAR(50) NOT NULL,   
  lastName VARCHAR(50) NOT NULL,   
  email VARCHAR(100) UNIQUE NOT NULL,   
  dateOfBirth DATE NOT NULL,   
  enrollmentDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,   
  psp DECIMAL(3, 2) CHECK (psp BETWEEN 0.00 AND 100.00),   
  batchId INT,   
  isActive BOOLEAN DEFAULT TRUE,   
  PRIMARY KEY (id),   
);
```

Annotations:

- auto-increment (points to `AUTO_INCREMENT`)
- not null (points to `NOT NULL`)
- varchar(50) (points to `VARCHAR(50)`)
- Watch Data Types video I shared (points to the `DECIMAL(3, 2)` and `PRIMARY KEY (id)` section)

Foreign Keys

Let's say we have years table as:

id	year	number_of_days
1	200	366
2	2001	365
3	2002	365
4	2003	365

and we have a table students as:

id	name	year
----	------	------

Is year column in students table a foreign key?

Note: Foreign key need not necessarily be the PK of the other table.

PK

id	year	no. of days.
1	200	366
2	2001	365
3	2002	365
4	2003	365

Students

(Default)

stud-id	first-name	last-name	batch-id
1	Mukul	Kumar	1
2	Raushik	Kumar	1
2	Kavya	Kannan	2
4	Sushmita	Ch	3
5	Durgesh	Khot	2

Annotations:
- Inconsistent: batch-id 1 is used for two different student IDs (1 and 2).
- Orphaning: student ID 4 is associated with batch-id 3, which is not a valid batch-id in the Batches table.

Batches

batch-id	batch-name
1	Batch A
2	Batch B
3	Batch C

→ In order to fix these problems with FK, MySQL allows to have constraints on operations like Delete and Update.

ON DELETE / ON UPDATE

- CASCADE
- SET NULL
- NO ACTION (Default action) (Most used)
- SET DEFAULT (only functional with tables that use InnoDB ENGINE)

CRUD
↓
Done → Read

Pseudo Code:-

answer = []

for each row in film:
 answer.append(row)

Select * from film;

[{?}, {?}, {?}, {?} ... - {?}]
 ↓ ↓ ↓ ↓
 row1 row2 row3.
 ↓
 row, id, des

filtered_answer = []

for each row in answer:

 filtered = answer.append(row['rating'], row['release-year'])

select rating, release-year from film;

unique_answer = set(filtered_answer);

return unique_answer.

select distinct rating, release-year
from film;

Insert Data from Another Table

Film



Film-copy



Insert data in this table

You need to create a
↑
table called film-copy.

Insert into film-copy (col names)

```
Values (
  (
  (
  (
```

```
select title, description, release-year . . . .
from Film;
```

title +

```
{
|
|
```